SM015
Mathematics 1
Semester I
Session 2023/2024
2 hours

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Matematik 1
Semester I
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2 jam

KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

S M 0 1 5

Kertas soalan ini mengandungi **14** halaman bercetak.
*This question paper consists of **14** printed pages.*

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of **3** questions. **Section B** consists of **7** questions.

Answer **all** questions.

All answers must be written in the answer booklet provided. Use a new page for each question.

Marks for each question or section are shown in the bracket at the end of the question or section.

All steps must be shown clearly.

The use of non-programmable scientific calculator is permitted.

Numerical answers may be given in the form of π, e, surd, simplest fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

ARAHAN KEPADA CALON:

*Kertas soalan ini mengandungi **Bahagian** A dan **Bahagian** B.*

***Bahagian** A mengandungi **3** soalan. **Bahagian** B mengandungi 7 soalan.*

*Jawab **semua** soalan.*

Semua jawapan hendaklah ditulis pada buku jawapan yang disediakan. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Markah yang diperuntukkan bagi setiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

Jawapan berangka boleh diberi dalam bentuk π, e, surd, pecahan termudah atau sehingga tiga angka bererti, di mana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin\left(A \pm B\right) = \sin A \cos B \pm \cos A \sin B$$

$$\cos\left(A \pm B\right) = \cos A \cos B \mp \sin A \sin B$$

$$\tan\left(A \pm B\right) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule
Petua Hasil Tambah

$$\frac{d}{dx}\left(u(x)+v(x)\right)=\frac{du}{dx}+\frac{dv}{dx}$$

Product Rule
Petua Hasil Darab

$$\frac{d}{dx}\left(u(x)\cdot v(x)\right)=v\frac{du}{dx}+u\frac{dv}{dx}$$

Quotient Rule
Petua Hasil Bahagi

$$\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right)=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$$

Chain Rule
Petua Rantai

$$\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{d}{dt}\left(\frac{dy}{dx}\right)\cdot\frac{dt}{dx}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Shape *Bentuk*	Volume *Isipadu*	Surface Area *Luas Permukaan*
Sphere *Sfera*	$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$
Right circular cone *Kon membulat tegak*	$V = \frac{1}{3}\pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder *Silinder membulat tegak*	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

SECTION A [25 *marks*]

***BAHAGIAN* A** [25 *markah*]

This section consists of **3** questions. Answer **all** questions.

Bahagian ini mengandungi **3** *soalan. Jawab* ***semua*** *soalan.*

1 Find

Cari

(a) $\lim_{x\to 2}\dfrac{10-3x-x^2}{x-2}$.

[3 *marks*]
[3 *markah*]

(b) $\lim_{x\to \infty}\dfrac{x+2}{x-1}$.

[3 *marks*]
[3 *markah*]

(c) $\lim_{x\to 3^-}\dfrac{|x-3|}{x^2-2x-3}$.

[4 *marks*]
[4 *markah*]

2 (a) Given

Diberi

$$f(x)=\begin{cases} x^2-3, & x<2, \\ \sqrt{x-1}, & x\geq 2. \end{cases}$$

Determine whether $f(x)$ is differentiable at $x=2$.

Tentukan sama ada $f(x)$ *boleh dibezakan pada* $x=2$.

[5 *marks*]
[5 *markah*]

(b) Differentiate $y=(x^2+1)\left(\sqrt{1-x^2}\right)$ with respect to x. Give your answer in the simplest form.

Bezakan $y=(x^2+1)\left(\sqrt{1-x^2}\right)$ *terhadap* x. *Berikan jawapan anda dalam bentuk teringkas.*

[4 *marks*]
[4 *markah*]

3 A liquid is to be cleared of sediment using a right circular-shaped cone filter. Assume that the height of the cone is 16 cm and the radius at the base of the cone is 4 cm. Given that the volume of the liquid is decreasing at a rate of 2 cm^3 per minute, find the rate of change in the liquid level when the depth of the liquid is 8 cm.

Suatu cecair perlu dibersihkan daripada mendapan dengan menggunakan penapis berbentuk kon membulat tegak. Andaikan bahawa ketinggian kon ialah 16 cm *dan jejari tapak kon ialah* 4 cm. *Diberi bahawa isipadu cecair berkurangan pada kadar* 2 cm^3 per minit, *cari kadar perubahan paras cecair apabila kedalaman cecair ialah* 8 cm.

[6 *marks*]
[6 *markah*]

SECTION B [75 *marks*]

***BAHAGIAN* B** [75 *markah*]

This section consists of **7** questions. Answer **all** questions.

Bahagian ini mengandungi **7** *soalan. Jawab* ***semua*** *soalan.*

1 Given that $-2m+(3+i)n=-4+3i$. Find the complex numbers m and n if n is the conjugate of m.

Diberi bahawa $-2m+(3+i)n=-4+3i$. *Cari nombor kompleks* m *dan* n *jika* n *adalah konjugat bagi* m.

[5 *marks*]
[5 *markah*]

2 (a) Given $\log_9\left(\log_{\sqrt{p}}\frac{1}{27}\right)=\frac{1}{2}$, find the value of p.

Diberi $\log_9\left(\log_{\sqrt{p}}\frac{1}{27}\right)=\frac{1}{2}$, *cari nilai bagi* p.

[4 *marks*]
[4 *markah*]

(b) Solve

Selesaikan

$$2^x+4<2^{2x}+2^x<12.$$

[8 *marks*]
[8 *markah*]

3 (a) Given

Diberi

$$f(x)=\frac{2e^{\frac{3}{2}x}}{e^{3x}-1} \; \begin{matrix} \textit{and} \\ \textit{dan} \end{matrix} \; g(x)=\frac{e^{3x}+1}{e^{3x}-1}.$$

Evaluate

Nilaikan

$$\left[f(x)\right]^2-\left[g(x)\right]^2.$$

[4 *marks*]
[4 *markah*]

(b) Given $g(x)=(x-2)^2-6$ and $h(x)=3x^2-12x+2$. Show that $f(x)=3x+8$, if $(f\circ g)(x)=h(x)$.

Diberi $g(x)=(x-2)^2-6$ *dan* $h(x)=3x^2-12x+2$. *Tunjukkan bahawa* $f(x)=3x+8$, *jika* $(f\circ g)(x)=h(x)$.

[4 *marks*]
[4 *markah*]

4 Given

Diberi

$$f(x)=2-\sqrt{3-x}.$$

(a) Find the domain and range of $f(x)$.

Cari domain dan julat bagi $f(x)$.

[2 *marks*]
[2 *markah*]

(b) Show that $f(x)$ is a one-to-one function by using algebraic approach.

Tunjukkan bahawa $f(x)$ *adalah fungsi satu ke satu dengan menggunakan pendekatan aljabar.*

[2 *marks*]
[2 *markah*]

(c) Determine $f^{-1}(x)$ and its domain.

Tentukan $f^{-1}(x)$ *dan domainnya.*

[4 *marks*]
[4 *markah*]

(d) Hence, sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes. Label your graph completely.

Seterusnya, lakarkan graf $f(x)$ *dan* $f^{-1}(x)$ *pada paksi yang sama. Labelkan graf anda selengkapnya.*

[3 *marks*]
[3 *markah*]

5 (a) Find the value of a if

Cari nilai bagi a jika

$$\lim_{x\to a}\frac{\sqrt{x}-2}{x-4}=\frac{1}{6}.$$

[3 *marks*]
[3 *markah*]

(b) The function $f(x)$ is given by

Fungsi $f(x)$ diberikan sebagai

$$f(x)=\begin{cases}1-2e^{2x}, & x<0,\\ 1 & x=0,\\ e^{3x}-2, & x>0.\end{cases}$$

(i) Determine the existence of the limit of $f(x)$ as x approaches 0.

Tentukan kewujudan had bagi $f(x)$ apabila x menghampiri 0.

[3 *marks*]
[3 *markah*]

(ii) Determine whether $f(x)$ is continuous at $x=0$. Hence, state the interval(s) where $f(x)$ is continuous.

Tentukan sama ada $f(x)$ adalah selanjar pada $x=0$. Seterusnya, nyatakan selang di mana $f(x)$ adalah selanjar.

[3 *marks*]
[3 *markah*]

5 (c) Given

Diberi

$$f(x) = \frac{\sqrt{x^3 + 5}}{x - 2}.$$

By finding the limit at appropriate values, obtain the horizontal and vertical asymptotes, if they exist.

Dengan mencari had pada nilai yang bersesuaian, dapatkan asimptot mengufuk dan menegak, jika ia wujud.

[5 *marks*]
[5 *markah*]

6 (a) Given

Diberi

$$xy - \sin(xy) - 3 = 0.$$

(i) Find $\frac{dy}{dx}$ in terms of x and y.

Cari $\frac{dy}{dx}$ *dalam sebutan* x *dan* y.

[4 *marks*]
[4 *markah*]

(ii) Hence, show that $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$.

Seterusnya, tunjukkan bahawa $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$.

[2 *marks*]
[2 *markah*]

(b) Given the parametric equations $x(t) = \ln(t^2 + t)$ and $y(t) = \sqrt{t^2 + t}$, for $t > 0$.

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t. Hence, evaluate $\frac{d^2y}{dx^2}$ when $x = \ln 2$.

Diberi persamaan berparameter $x(t) = \ln(t^2 + t)$ *dan* $y(t) = \sqrt{t^2 + t}$, untuk $t > 0$. *Cari* $\frac{dy}{dx}$ *dan* $\frac{d^2y}{dx^2}$ *dalam sebutan* t. *Seterusnya, nilaikan* $\frac{d^2y}{dx^2}$ *apabila* $x = \ln 2$.

[10 *marks*]
[10 *markah*]

7 Given

Diberi

$$f(x) = \cos x + \sin x,\ 0 \leq x \leq 2\pi.$$

(a) By using the first derivative test, find the extremum points.

Dengan menggunakan ujian terbitan pertama, cari titik-titik ekstremum.

[5 *marks*]
[5 *markah*]

(b) Determine the inflection points of $f(x)$.

Tentukan titik-titik lengkuk balas bagi $f(x)$.

[4 *marks*]
[4 *markah*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT